



DCSENNets: Interpretable deep learning for patient-independent seizure classification using enhanced EEG-based spectrogram visualization[☆]

Sunday Timothy Aboyeji^{a,b,c,1}, Ijaz Ahmad^{a,c,1}, Xin Wang^{a,c,1}, Yan Chen^d, Chen Yao^e, Guanglin Li^{a,c}, Michael Chi Fai Tong^f, Alice K.Y. Siu^{g,1,*}, Guoru Zhao^{a,c,*}, Shixiong Chen^{g,1,*}

^a CAS Key Laboratory of Human–Machine Intelligence Synergy Systems, Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen, Guangdong 518055, China

^b University of Chinese Academy of Sciences, Beijing 100049, China

^c Guangdong-Hong Kong-Macao Joint Laboratory of Human–Machine Intelligence-Synergy Systems, Shenzhen, Guangdong 518055, China

^d Surgery Division, Epilepsy Center, Shenzhen Children's Hospital, Shenzhen 518036, China

^e Department of Neurosurgery, The National Key Clinic Specialty, Shenzhen Key Laboratory of Neurosurgery, the First Affiliated Hospital of Shenzhen University, Shenzhen Second People's Hospital, Shenzhen 518035, China

^f Department of Otorhinolaryngology, Head and Neck Surgery, The Chinese University of Hong Kong, Hong Kong, China

^g School of Medicine, The Chinese University of Hong Kong, Shenzhen, Guangdong 518172, China

ARTICLE INFO

Keywords:

EEG
Epileptic seizure
Enhanced spectrograms visualization
Taper functions
Dilated convolutional squeeze and excitation networks
Kolmogorov–Smirnov test

ABSTRACT

Neurologists often face challenges in identifying epileptic activities within multichannel EEG recordings, requiring extensive hours of analysis. Computer-aided diagnosis systems have been proposed to reduce manual inspection of EEG signals by neurologists. However, direct analysis of EEG signals is difficult due to their complex and dynamic nature, with variation across multiple patients. Therefore, researchers have proposed the short-time Fourier transform (STFT) to capture dynamic events indicative of seizures through time-varying frequency representation of EEG signals. However, tradeoffs between time and frequency resolution limited the spectrogram's interpretability and affected clinical deployment. Hence, this study proposes extracting high-resolution channels via a novel STFT spectrogram construction algorithm encompassing taper functions for seizure diagnosis. Initially, we extracted seizure and non-seizure segments from each channel of selected patients in the CHB-MIT dataset. Next, we systematically apply taper functions like Hann and Gaussian windows to minimize the edge effect during the construction of spectrogram images. Finally, we employ Dilated Convolutional Squeeze and Excitation Networks (DCSENNets) through leave-one-patient-out cross-validation (LOPOCV) to perform patient-independent seizure classification. The proposed DCSENNets achieve an average accuracy of $87.20 \pm 11.48\%$ and $87.29 \pm 10.48\%$ with Hann and Gaussian taper functions, respectively, and $86.85 \pm 11.56\%$ without the taper function. Most patients with high performances indicate similarity in train-test sample distribution using the Kolmogorov–Smirnov test at $0.01 < p \leq 0.05$ or $p > 0.05$. Furthermore, the Grad CAM deep visual explainer integration enhances the interpretability of the deep learning model's decision-making process. Consequently, neurologists are provided not only with enhanced visualized spectrograms but also a transparent model for improved seizure diagnosis.

1. Introduction

Epilepsy is a neurological disorder characterized by repeated seizures, which are the result of sudden bursts in electrical activities

in the brain that affect its standard functionality. Seizures restrict individuals from taking on specific responsibilities. The condition requires continuous management through a proper diagnosis process.

[☆] This work was supported in part by the National Key R&D Program of China (#2022YFE0197500), National Natural Science Foundation of China #62471422, #62401558), Shenzhen Medical Research Fund (#D2402003), Science and Technology Planning Project of Shenzhen (#JCYJ20230807093819039), China Postdoctoral Science Foundation (#2024M753402), and the Postdoctoral Fellowship Program of CPSF under Grant Number (#GZC20241849).

* Corresponding authors.

E-mail addresses: timothy@siat.ac.cn (S.T. Aboyeji), ijaz@siat.ac.cn (I. Ahmad), wangxin@siat.ac.cn (X. Wang), ccflysz@126.com (Y. Chen), pro.yaochen@gmail.com (C. Yao), gl.li@siat.ac.cn (G. Li), mtong@ent.cuhk.edu.hk (M.C.F. Tong), alicesiu@cuhk.edu.cn (A.K.Y. Siu), gr.zhao@siat.ac.cn (G. Zhao), chenshixiong@cuhk.edu.cn (S. Chen).

¹ Contributed equally to this work.

Seizures are usually monitored through neurological exams, blood tests, neuropsychological tests, or neuroimaging [1] by using devices such as magnetic resonance imaging (MRI), computed tomography (CT), positron emission tomography (PET), and electroencephalogram (EEG) [2]. However, EEG is most preferred because of its portability, cost-effectiveness, high temporal resolution, and essentiality for differentiating types of seizures. For diagnosis, neurologists usually monitor the differences in the wave patterns of the EEG recording simultaneously across multiple channels. However, Channels closer to the epileptogenic zone show significant amplitude variation compared to channels farther away.

In addition, epileptic seizures can be classified into partial and generalized seizures. Partial seizures start from a single spot on the brain and are thought to have a focal onset. In contrast, generalized seizures start simultaneously in both hemispheres and are believed to have a generalized onset. Therefore, we cannot over-emphasize the contribution of individual channels to seizure detection, particularly for patients with focal seizures. Recently, machine and deep learning models have gained significant popularity in computer-aided diagnosis systems (CADs) for epileptic seizure detection (ESD). Moreover, deep learning models like convolutional neural networks (CNNs), long short-time memory (LSTM), recurrent neural networks (RNNs), and autoencoders (AEs) have gained fame for ESD because they can directly extract feature maps, unlike machine learning models that require feature engineering. CNNs and RNNs can precisely capture important temporal and spatial features in EEG signals [3].

A one-dimensional CNN model with noise removal discrete wavelet transform (DWT) was used by [4] to perform patient-independent seizure prediction. Wang et al. [5] combine CNN and LSTM models' architectures to achieve a spatiotemporal network for ESD. In their approach, CNN was used to extract the spatial features, while LSTM captured the time-series information of EEG signals. In another study, Samee et al. [6] used a hybrid of RNN and BiLSTM to perform automatic ESD based on EEG signals. However, Shoka et al. [7] suggested that converting EEG signals to spectrogram images would allow DL models to extract meaningful features by focusing on relevant channels. Furthermore, the introduction of a channel attention mechanism for ESD by [8] allows more focus to be placed on specific channels in the feature maps of the deep learning networks. They used 3D CNN in their study to extract features from the spectrograms of EEG signals. After that, they introduced a convolutional block attention module (CBAM) to capture spatial and channel attention for seizure prediction. However, the generated spectrograms do not clearly show power spectrums differentiated at each frequency component, which could limit clinical application. Also, this could hinder the interpretation of spectrograms for precise clinical decision-making for ESD.

Additionally, Ra et al. [9] emphasized the importance of obtaining higher-resolution short-time Fourier transform (STFT) images for seizure classification. Hence, they proposed Synchroextracting Transformation (SET) and Singular Value Decomposition (SVD) to improve the time-frequency resolution. This approach is commendable but computationally intensive, as complex algorithms involving decomposition into large matrices are involved, which could limit its real-time application. Moreover, SET-SVD optimized on a particular patient's dataset might not translate to other patients limiting its generalization ability for patient-independent ESD. Hence, the authors in [8] applied STFT directly to the segments of the multichannel EEG signal. This approach complicates the clinical interpretation of individual channels and results in significant information loss. To concentrate on specific channel information, Handa and Goel [10] constructed spectrograms for separate channels of EEG signals. However, their approach pays less attention to edge effects while constructing the spectrograms. An edge effect or artifact is a distortion in the spectral representation of a time series signal caused by fluctuation in the initial and final values of the empirical input series [11]. Therefore, this study introduces two distinct taper functions to account for the edge effect in the EEG signal.

Furthermore, the previous studies paid little attention to time-frequency resolution when constructing the spectrograms to enhance the visual discrimination of channels for effective clinical diagnosis of seizures. Hence, we have considered different window sizes while constructing the spectrogram images. Moreover, the window size for the signal segmentation when applying STFT can significantly influence the visualization of the spectrograms [12]. Therefore, our channel-based STFT allows focus to be placed on individual channels from different patients when constructing spectrograms.

Additionally, the distribution of the power spectrum density (PSD) in the time-frequency spectrograms can influence the nature of discriminating features extracted by the DL models. Therefore, we have introduced squeeze and excitation (SE) blocks between the dilated convolutional neural network to pay attention to respective channels in the feature maps and efficiently differentiate between seizure and normal spectrogram images. Precisely, the channel-wise attention based on the SE networks can model the interdependencies between channels of convolutional features [13]. This enhances the generalization and improves the interpretability of CNNs for seizure classification. Therefore, we have conducted experiments on publicly available CHB-MIT datasets for patient-independent seizure classification using Dilated Convolutional Squeeze and Excitation Networks (DCSENNets). In summary, the main contributions of this paper are:

- Provides taper functions such as Hann and Gaussian with different window sizes incorporated into an enhanced spectrogram visualized algorithm to preprocess EEG signals, accounts for edge effects, and ensures effective differentiation of time-frequency components.
- Presents dilated convolutional squeeze and excitation network (DCSENNets) to extract and focus attention on essential features in the channels for improved ESD.
- Guarantee generalized patient-independent seizure classification through leave-one-patient-out cross-validation (LOPOCV) evaluation technique and iterative model performance optimization algorithm.
- Executes Kolmogorov–Smirnov statistical test at a threshold of $0.01 < p \leq 0.05$ or $p > 0.05$ to evaluate training and test sample distribution in LOPOCV implementation.
- Suggests explainable AI (XAI) using gradient-weighted class activation mapping (Grad-CAM) to visualize and understand the decision-making process of deep learning model on the spectrograms for ESD leading to improved clinical interpretability.

2. Methodology

Fig. 1 shows the proposed patient-independent seizure detection framework. The approach involves multichannel EEG data preprocessing and transformation into spectrogram images before being classified using DCSENNets. In addition, the section describes various taper functions that enhance spectrogram construction, as well as the proposed optimization algorithm incorporated into DCSENNets following the LOPOCV evaluation approach.

2.1. Data description and preprocessing

This study adopted the publicly available CHB-MIT datasets from Boston Children's Hospital, United States. The dataset contains multichannel EEG recordings from 22 pediatric subjects with intractable seizures [14]. The recordings were acquired at 256 samples per second with 16-bit resolution. The bipolar montage international 10–20 channel configuration was adopted to acquire the dataset. Initially, noise and other artifacts are removed using a finite impulse response (FIR) filter through a Hamming window. The filter is characterized by a passband ripple of 0.0194 and stopband attenuation of 53 dB, which effectively allows frequency passage between 0.5 Hz and 30 Hz.

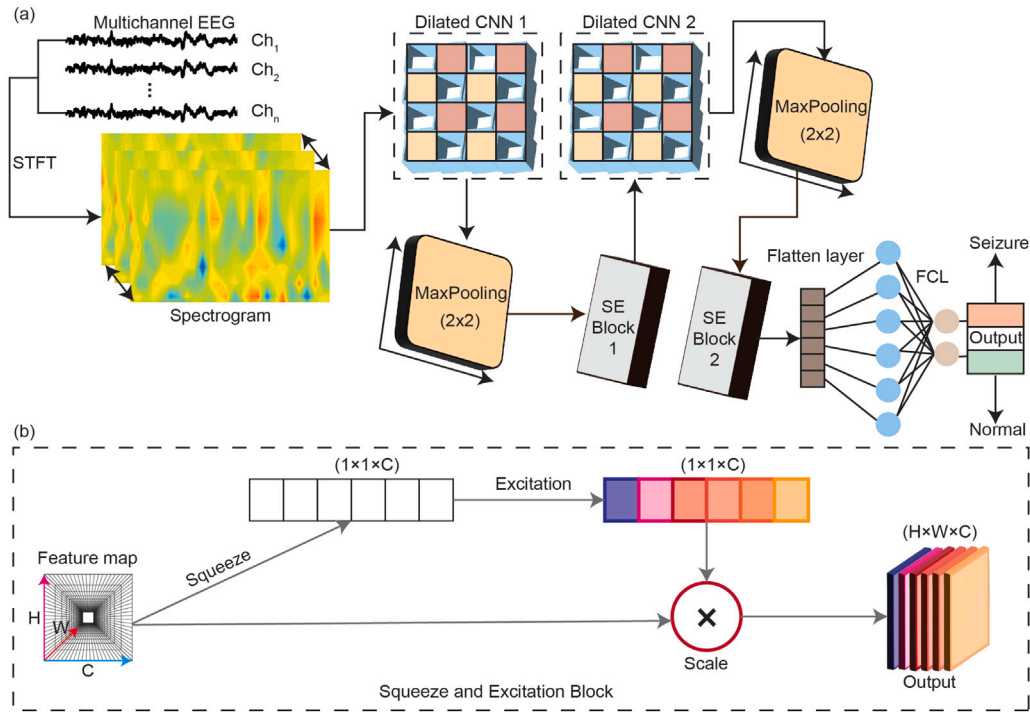


Fig. 1. Framework of the proposed dilated convolution squeeze and excitation network. (a) The complete structure involves the conversion of multichannel EEG signals to spectrogram images, followed by the operation of dilated convolutional networks with max-pooling layers, and then squeeze and excitation (SE) blocks. The flattened layer and the fully connected layer (FCL) are added to convert the multidimensional output into a single vector for classification into seizure and normal. (b) The SE blocks perform squeeze, excitation, and scaling operations to focus attention on the informative channel in the feature maps.

Table 1
Details of the patients with number of constructed spectrogram images.

Patient ID	Gender-Age (years)	Seizure event	Total seizure duration (s)	STFT images
Chb01	F - 11	7	442	154
Chb02	M - 11	3	272	66
Chb03	F - 14	7	402	154
Chb04	M - 22	4	378	88
Chb05	F - 7	5	860	110
Chb06	F - 1.5	9	137	198
Chb07	F - 14.5	3	325	66
Chb08	M - 3.5	5	919	110
Chb09	F - 10	4	276	88
Chb10	M - 3	7	447	154
Chb16	F - 7	5	47	110
Chb17	F - 12	3	294	66
Chb18	F - 18	6	297	132
Chb19	F - 19	3	236	66
Chb20	F - 6	8	294	176

Note: Equivalent intervals of seizure duration are obtained for the normal class, generating the same number of images from those segments.

Thereafter, 22 channels namely FP1-F7, F7-T7, T7-P7, P7-O1, FP1-F3, F3-C3, C3-P3, P3-O1, FP2-F4, F4-C4, C4-P4, P4-O2, FP2-F8, F8-T8, T8-P8, P8-O2, FZ-CZ, CZ-PZ, P7-T7, T7-FT9, FT9-FT10, and FT10-T8. The repeated T8-P8, the last channel, has been removed since two of this channel were in the recordings [2]. Hence, we constructed the spectrograms using datasets from 15 patients, effectively representing the age and gender distribution of the patient population (Table 1). The numbers of generated images for each patient using the 22 channels for each recorded seizure event are shown in Table 1. Similarly, normal spectrogram images corresponding to the total number of seizure events are generated from the interictal period of the recordings. Daoud and Bayoumi [15] suggest that the interictal phase should occur at least four hours before or after a seizure for accurate prediction. However, because our work focuses on detection rather than prediction, the

interictal phase does not have to be as far apart from the preictal time. As a result, we proposed normal intervals of at least 2 h before or after a seizure.

2.2. Window (taper) function and STFT

Two essential taper functions are applied in this study. EEG signals are known to be nonlinear and nonstationary and vary significantly, especially during seizure events. Applying taper functions helps minimize the edge effects (large variation at the end and beginning of time-series data leading to spectral distortion). The taper function usually ramps time-series data from zero to one and back to zero, thereby weighting the data series in a way that forces the ends of the data vector to zero [11]. The choice of taper functions depends on how they ramp the time-series signal. Therefore, Hann and Gaussian window functions are considered in this study because of their popularity in dealing with edge artifacts.

2.2.1. Hann (Hanning) taper function

The Hann taper function (HTF), invented by Julius von Hann around 1900, can be used to tape a signal [16]. The Hanning window is frequently preferred for analyzing random signals [17] because of its ability to reduce random errors. The Hann window described by (1) is a taper function that smoothens the edges of the signal.

$$w[n] = \frac{1}{2} \left[1 - \cos \left(\frac{2\pi n}{N} \right) \right], \quad 0 \leq n \leq N \quad (1)$$

where N is the total number of samples, which can be even or odd. In this case, the amplitude is 1, and the sample number N is set to the EEG signal's sampling frequency (256 Hz).

2.2.2. Gaussian taper function

Gaussian taper function (GTF) has been extensively applied for STFT analysis in signal processing for optimal time-frequency resolution [18]. Shimauchi et al. [19] further experiment with the importance of GTF when using STFT to establish a relationship between amplitude

and phase. They explained that the instantaneous frequency of each signal segment derived from the phase definition can be explicitly linked with the amplitude without depending on signal models. The GFT is defined by

$$w[n] = \exp\left(-\frac{1}{2} \left(\frac{n - N/2}{\sigma N/2}\right)^2\right), \quad 0 \leq n \leq N. \quad (2)$$

where n is the typical value at the sample index, N is the total length of the window (sample frequency), and the expression $\sigma N/2$ is the standard deviation. The variable σ makes the GTF a flexible window function [20]. Moreover, this parameter controls the spread of the window, with the smaller value of σ producing a narrow window. Selected values of σ influence the balance between the width of the main lobe and the levels of the side lobes in frequency domain analysis, which in turn affects the precision and the amount of leakage in spectral estimates [21]. Therefore, in this study, five different values of σ are considered, as shown in Fig. 2.

2.2.3. Short time Fourier transform

The short-time Fourier transform (STFT) can be described as a sequence of Fourier transforms of a windowed signal [22]. The continuous STFT of a continuous time-domain function, x_τ is given as:

$$X_{t,\omega} = \int_{-\infty}^{\infty} w_{\tau-t} x_\tau e^{-j\omega(\tau-\frac{T}{2})} d\tau, \quad (3)$$

where $w_{\tau-t}$ is the sliding taper function (which is either HTF or GFT in this study) that concentrates the signal around time $\tau = t$, and $X_{t,\omega}$ is the complex-valued output of STFT which is a function of time t and angular frequency ω [19]. The discrete Fourier transform window length is represented by T . The process by which the number of spectrogram (STFT) images mentioned in Table 1 are generated is illustrated in Algorithm 1. First, we obtain the time point within the specified window length. For each of the channels, the data is segmented according to the calculated time points to the defined time window around the time point. The taper functions, such as Hann and Gaussian, are then applied to reduce the spectral distortion (Fig. 3b–c). The power spectrum was then computed from the fast Fourier transform of the tapered signal (Fig. 3d). The obtained power spectrum for each segment is then interpolated over the predefined frequency range. The resulting outputs are normalized and converted to decibels (dB) to improve visualization of the spectrogram in time-frequency representation. Fig. 3(e) and (f) show the contour plot (spectrograms) of the time-frequency representation of normal and seizure segments of EEG signal for typical channels (F3–C3), respectively. As shown in the figure, the channel with a seizure event exhibits a high power distribution in the time-frequency representation based on color differentiation. In addition, the power range of normal (−16.0 to 9.6 dB) and seizure (−36 to 12 dB), as shown in the color bar, indicates the variation in the signal power intensity during these states.

2.3. Dilated convolutional neural networks

Convolutional neural networks (CNNs) are the earliest deep learning models that have been extensively applied in medical image segmentation, computer vision, voice recognition, and object detection [23]. CNNs are particularly important when extracting feature maps from spectrogram images of EEG signals [8,10]. Moreover, 2DCNN can learn discriminating features from multisource data at high resolution to enhance generalization in the detection algorithm [24]. Considering each pixel of the spectrogram $x_{(m,n)}$, the output of the operation of 2DCNN is given by (4) [25].

$$y_{mn} = f\left(\sum_{i=0}^{h-1} \sum_{j=0}^{w-1} k_{ij} x_{(m+i)(n+j)} + \text{bias}_{mn}\right) \quad (4)$$

where k_{ij} is the 2D convolutional kernel of size $(h \times w)$ that extracts features from the spectrogram images. Initial generated images through Algorithm 1 are of size 462×775 but resized to 224×224 , which is

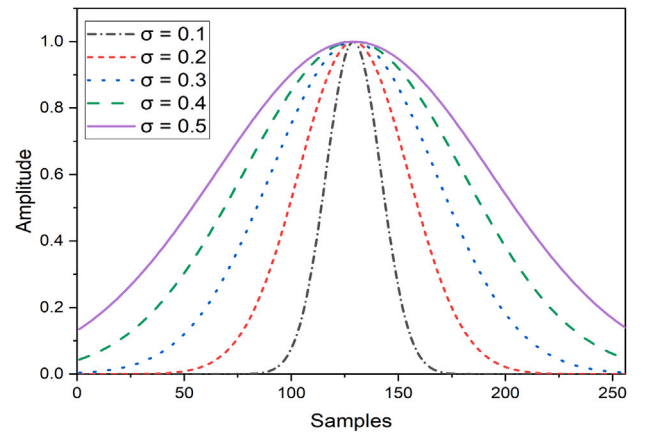


Fig. 2. Gaussian taper function illustrating different sigma values that control the spread of the window.

Algorithm 1 Enhanced STFT spectrogram visualization

Require: EEG data with channels (Ch)

Ensure: Spectrogram images

- 1: Initialization: $WL \leftarrow$ set time window, $TR \leftarrow$ set time range, $F_s \leftarrow 256$
- 2: $F \leftarrow$ log spacing between 0.5 and 40 Hz
- 3: $T_{idx} \leftarrow$ get the time indices within the TR
- 4: $TWS \leftarrow WL \times F_s$ ▷ setting time window samples
- 5: $WF \leftarrow$ set the window function using (1), (2) and (3)
- 6: $AllCh \leftarrow$ Zero matrix of shape $(Ch, \text{len}(F), \text{len}(T_{idx}))$
- 7: $STFT \leftarrow$ Zero matrix of shape $(\text{len}(F), \text{len}(T_{idx}))$
- 8: **for** $i \leftarrow 1$ to Ch **do**
- 9: **for** $j \leftarrow 1$ to $\text{length}(T_{idx})$ **do**
- 10: $StartIdx \leftarrow \max(T_{idx}[j] - TWS/2, 0)$
- 11: $EndIdx \leftarrow \min(T_{idx}[j] + TWS/2, N)$
- 12: $Data_{seg} \leftarrow$ EEG data[$i, StartIdx : EndIdx$]
- 13: $Data_{tapered} \leftarrow Data_{seg} \times WF$
- 14: $y \leftarrow \text{FFT}(Data_{tapered}, TWS)$ ▷ $TWS/2 + 1$
- 15: $P \leftarrow |y|^2$ ▷ compute the power spectrum
- 16: $STFT[:, j] \leftarrow$ interpolate P at defined frequencies
- 17: **end for**
- 18: $AllCh[i, :, :] \leftarrow 10 \log_{10}(STFT / \text{mean}(STFT))$
- 19: **end for**
- 20: Spectrogram images $\leftarrow$ generate spectrogram images
- 21: **return** Spectrogram images

the famous size for computer vision and image processing applications. The activation function $f()$ is set to ReLU as in (5).

$$f(x) = \max(0, x) \quad (5)$$

ReLU is the most prominent activation function in a neural network, especially CNN, as it overcomes the vanishing gradient problem in sigmoid and tanh.

The CNN architecture often requires a higher receptive field when processing significant spatial contexts such as spectrograms to capture long dependencies. Increasing the network's deepness and kernel size would increase the receptive field but introduce complexity to the model. Therefore, introducing a dilated convolution would enable the network to function at a broader scale comparable to standard convolution with a larger kernel size without increasing the number of training parameters [26]. The dilated convolution neural network (Dil. CNN) with the rate r increases the size of k filter to $[k + (k - 1)(r - 1)] \times [k + (k - 1)(r - 1)]$ by the introduction of $r - 1$ factor [27]. In this study, the dilation rate of size (2×2) is applied to enlarge the receptive field of the convolution network.

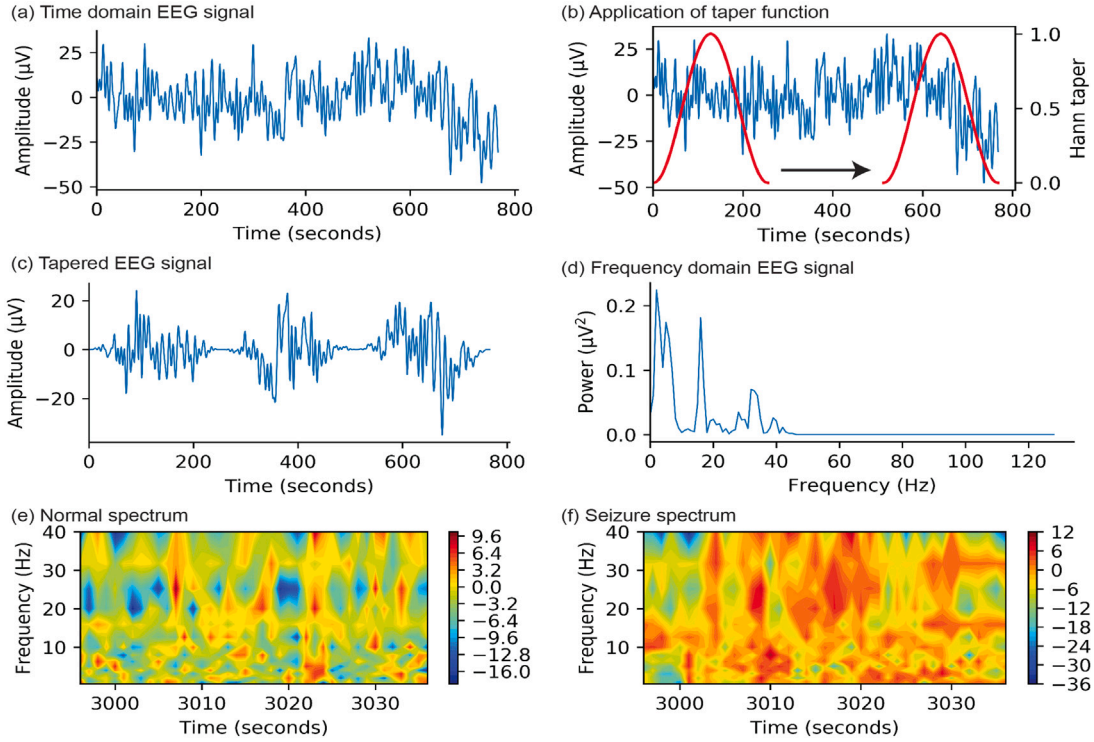


Fig. 3. The filtered EEG signal in the time domain is illustrated by (a). The application of the Hann taper function to reduce the edge effect by (b). The tapered EEG signal with removed edge effect is illustrated by (c). The power spectrum obtained in the frequency domain for a typical EEG signal by (d). The spectrograms of the normal and seizure recordings from channel F3-C3 representing seizure-free and seizure events for a period of 40 s are (e) and (f), respectively.

2.4. Squeeze and excitation network

Squeeze-and-excitation networks (SENet), first proposed by Hu et al. can be used to scale channel-wise feature maps by explicitly paying attention to inter-dependencies between channels [28]. Since its introduction, SE has gained significant popularity in enhancing the importance of informative features within the feature map and suppressing the non-essential ones. SE blocks are usually integrated with commonly used deep learning frameworks to improve their feature representation capacity. In [13], the SE-ResNet module was introduced for epileptic seizure detection by combining the SE block with the residual neural network (ResNet) block. The approach employed a channel-weighted model, paying attention to the contribution of each channel. The ability of SE blocks to be combined with any CNN architecture makes them useful in image classification, especially for the spectrograms of the EEG signals considered in this study. The operation of the SE block can be divided into squeeze, excitation, and scaling parts, as shown in Fig. 1(b). According to [28], the learned filters operate with a local receptive field and cannot capture similar patterns outside this field. Therefore, the squeeze operation in the SE block first performs global average pooling by aggregating the feature maps output U across spatial dimensions ($H \times W$) to produce channel descriptor $z \in \mathbb{R}^C$ as in (6) [28].

$$z_c = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W u_c(i, j) \quad (6)$$

The excitation operation involves recalibrating the aggregated information in the preceding operation. A simple gating mechanism is implemented through a sigmoid activation function involving two fully connected (FC) layers, as in (7).

$$s = \sigma(W_2, \delta(W_1 z)) \quad (7)$$

where σ and δ are the sigmoid and ReLU activation functions, respectively, $W_1 \in \mathbb{R}^{\frac{C}{r} \times C}$ and $W_2 \in \mathbb{R}^{C \times \frac{C}{r}}$ are the weights of the FC layers and

Table 2

Layer parameters of DiL.CNN and DCSENet.

Layer name	Output shape	DiL.CNN	DCSENet
Input Layer	(224, 224, 3)	0	0
Sequential	(224, 224, 3)	0	0
Conv2D	(220, 220, 32)	896	896
2D MaxPooling	(110, 110, 32)	0	0
SE block	(110, 110, 32)	–	128
Conv2D	(106, 106, 32)	9248	9248
2D MaxPooling	(53, 53, 32)	0	0
SE block	(53, 53, 32)	–	128
Dropout	(53, 53, 32)	0	0
Flatten	(89888)	0	0
Dense	(64)	5752.896	5752.896
Dense	(1)	65	65
Total parameters		5,763,105	5,763,361

r is a constant factor [29]. The first FC performs channel dimensionality reduction to capture the channel-wise relationship, while the second increases the dimension to match the original feature maps. Finally, the scaling operation is performed on the excitation operation's output (set of channel-wise weights). The output is the multiplication operation between the scaler S_c and feature map u_c as in (8) [28]. In addition, the SE blocks we added between the convolutional neural networks in our proposed model can adaptively recalibrate feature responses.

$$\tilde{X}_s = s_c \cdot u_c \quad (8)$$

In this way, the proposed dilated convolutional squeeze and excitation networks (DCSENet) can focus on essential features in the EEG spectrograms for seizure detection. The SE block is computationally efficient for improving model performance with the addition of a few more parameters, as shown in Table 2.



Fig. 4. Patient-independent LOPOCV seizure detection evaluation procedure for the proposed model. For each loop, the model is trained on $n - 1$ patients and tested on one patient entirely not part of the training samples.

2.5. Leave one patient out cross validation

This study implements a patient-independent seizure detection approach using leave-one-patient-out cross-validation, as shown in Fig. 4. The LOPOCV is now gaining attention to evaluate the performance of deep learning for epileptic seizure prediction [30,31] and detection [32]. In the former, the leave-one-out cross-validation involves leaving specific seizure event files in a patient-specific approach (LOPOCV) while training the others. However, the latter approach adopted in this study is entirely based on leaving one patient out to test the model in each fold. As shown in Fig. 4, $n - 1$ patients are trained for each fold, and testing is done on the left out patient. This procedure ensures that the model is entirely blind to the test dataset. By this, we aim to ensure a robust patient-independent model with high generalization capability.

3. Experimental results

3.1. Experimental set-up and evaluation metrics

The experiments of this study performed on the CHB-MIT datasets were carried out on an Intel(R) Xeon(R) E5-2630 CPU with installed RAM of 128 GB and a speed of 2.10 GHz. The execution used Python 3 with packages such as NumPy for numerical computation, Matplotlib for visualization, glob for file exploration, Magnetoecephalography and Electroencephalography (MNE) for reading and manipulating edf files, Python Imaging Library (PIL) for image transformation, TensorFlow version 2.3.0 to train the model and scikit-learn for model evaluation on windows 10 with 64-bit operating system. A fast training process for the LOPOCV approach is achieved by using the available NVIDIA TITAN Xp GPU with 75.9 GB memory. A batch size of 32 and an Adam optimizer with a learning rate of 0.001 for 20 epochs are applied in training the model. Initially, we applied data augmentation techniques, including random horizontal flips, random rotations, and random zooms, to generate modified versions of our training sets due to the limited size of the spectrogram images. Additionally, data augmentation helps prevent overfitting and improves the generalization of the patient-independent model.

Furthermore, we optimized the proposed model's performance using an iterative optimization algorithm. In the algorithm, the training datasets are randomly shuffled to prevent overfitting and improve generalization to unseen datasets. Each training epoch randomly combines shuffled samples from different patients into batches. The algorithm then finds the optimal model weights by evaluating the best combination of samples from P_{n-1} patients' datasets on the P_i test patient, as detailed in Algorithm 2.

Three essential metrics (acceleration, sensitivity, and specificity) evaluate the model's performance. Accuracy mathematically described by (9) is the ability of the model to make correct classification over the entire spectrogram samples [8,30]. Sensitivity, also known as Recall or

True Positive Rate, measures the ability of the model to identify seizure patterns in the spectrogram accurately [8]. In contrast, specificity, known as True negative rate, is the ability of the model to determine the normal pattern in the spectrograms [30].

$$\text{Accuracy} = \frac{T_p + T_n}{T_p + T_n + F_p + F_n} \quad (9)$$

$$\text{Sensitivity} = \frac{T_p}{T_p + F_n} \quad (10)$$

$$\text{Specificity} = \frac{T_n}{T_n + F_p} \quad (11)$$

where T_p (true positive) represents the number of spectrogram images correctly identified as seizures, T_n (true negative) represents the number of spectrogram images correctly identified as normal, F_p (false positive) represents the number of non-seizure spectrogram images incorrectly identified as seizures, and F_n (false negative) represents the number of seizure spectrogram images incorrectly identified as normal by the model.

Algorithm 2 Iterative model performance optimization

Require: Training and test samples

Ensure: Best model

```

1: Initialization: Accuracy  $Acc = -\infty$ , Epochs  $E = 20$ 
2:  $(X_{\text{train}}, y_{\text{train}}) \leftarrow P_1, P_2, \dots, P_{i-1}, P_{i+1}, \dots, P_n$   $\triangleright$  Train samples
3:  $(X_{\text{test}}, y_{\text{test}}) \leftarrow P_i$   $\triangleright$  Test samples
4: for  $n \leftarrow 1$  to  $E$  do
5:    $CurrentModel \leftarrow \text{train model}(X_{\text{train}}, y_{\text{train}})$ 
6:    $Score \leftarrow \text{evaluate model}(X_{\text{test}}, y_{\text{test}})$ 
7:   if  $Score > Acc$  then
8:      $Acc \leftarrow Score$ 
9:      $BestModel \leftarrow \text{save the weights of } CurrentModel$ 
10:  end if
11: end for
12: return  $BestModel$ 

```

3.2. Ablation study

3.2.1. Filter size

Initially, we performed an ablation study based on filter sizes of 3×3 , 5×5 , and 7×7 , as shown in Fig. 5. The results indicate that the filter sizes $k = 3$ and $k = 5$ show similar performances with an average accuracy of 87.20% and 86.93%, respectively, for complete LOPOCV training. However, the filter size of $k = 7$ reduces the model's performance, as indicated in Fig. 5, with an average accuracy of 80.37% achieved. Furthermore, when $k = 3$, the computational demand is less than when $k = 5$, and much less when $k = 7$. Therefore, we have adopted a $k = 3$ filter size for our patient-independent seizure detection, as the model can capture small localized patterns with a focus on the textures and edges of the features.

3.2.2. Gaussian function variable

As shown in Fig. 2, the spread of the Gaussian window can be controlled by different sigma values. First, an ablation study was performed to find the suitable sigma values when three different window lengths (1 s, 5 s, and 10 s) were used. Fig. 6 illustrates the performance of GTF across different sigma values considering the window lengths. It can be observed that GTF exhibits superior performance at $\sigma = 0.3$ for a window length of 1 s (WL = 1 s). However, when the window length is increased to WL = 5 s and WL = 10 s, GTF achieves optimum performance at $\sigma = 0.1$ as indicated by the accuracy scores and standard deviation values. Therefore, these WLs with their corresponding sigma values are used when computing GTF for the proposed model.

Table 3

The performance of proposed DCSENet model for individual patient in LOPOCV approach.

Patient ID	Train	Test	Hann taper function				Gaussian taper function				Without taper function			
			WL = 1 s	WL = 5 s	WL = 10 s	Avg p-value	WL = 1 s	WL = 5 s	WL = 10 s	Avg p-value	WL = 1 s	WL = 5 s	WL = 10 s	Avg p-value
Chb01	3168	308	86.69	79.55	80.84	0.3685	87.34	86.36	82.47	0.0917	84.42	82.14	82.47	0.3123
Chb02	3344	132	87.88	78.03	73.48	0.0177	87.88	87.12	82.58	0.0905	82.58	78.03	65.91	0.0399
Chb03	3168	308	92.53	90.58	88.31	0.0353	93.18	90.58	92.21	0.0000	92.21	92.53	90.26	0.1621
Chb04	3300	176	84.66	82.39	85.23	0.0015	84.09	82.95	82.95	0.0009	83.52	84.66	82.39	0.0000
Chb05	3256	220	91.36	91.36	99.09	0.0139	93.18	93.18	90.00	0.3202	92.73	93.18	96.82	0.0005
Chb06	3080	396	59.60	54.29	50.76	0.0003	61.61	57.83	57.58	0.0003	58.08	52.02	49.75	0.0002
Chb07	3344	132	99.24	99.24	98.48	0.3273	98.48	98.48	99.24	0.4173	97.73	98.48	98.48	0.1264
Chb08	3256	220	76.82	72.73	75.00	0.0000	76.82	78.18	79.09	0.0000	73.18	66.82	70.45	0.0018
Chb09	3300	176	98.86	99.43	97.73	0.0072	97.16	100.00	100.00	0.0221	99.43	97.73	98.30	0.0290
Chb10	3168	308	91.23	87.99	89.29	0.1616	91.88	91.56	89.61	0.0001	89.29	90.26	92.53	0.0312
Chb16	3256	220	64.09	64.55	62.73	0.0000	67.72	63.18	63.18	0.0000	69.55	59.09	62.27	0.0000
Chb17	3344	132	89.39	90.15	88.64	0.0000	90.15	88.64	88.64	0.0000	91.67	90.91	83.33	0.0001
Chb18	3212	256	94.32	90.91	88.26	0.4591	89.77	94.70	96.21	0.2114	95.83	91.67	89.02	0.2385
Chb19	3344	132	99.24	100.00	100.00	0.0263	99.24	100.00	100.00	0.0014	100.00	100.00	100.00	0.0288
Chb20	3124	352	92.05	92.05	88.07	0.0009	90.91	92.05	91.48	0.0036	92.61	92.61	90.91	0.0070
Average accuracy			87.20	84.88	84.39		87.29	86.99	86.35		86.85	84.68	83.53	
Standard deviation			11.48	12.71	13.52		10.48	11.95	12.07		11.56	14.15	14.55	

The accuracy scores are in percentage. The average p-values are obtained from the distribution between train and test samples using the Kolmogorov–Smirnov test across the three window lengths. The highlighted p-values indicate where $(0.01 < p \leq 0.05 \text{ or } p > 0.05)$, suggesting no significant difference between the test samples of that patient and the trained samples of 14 patients.

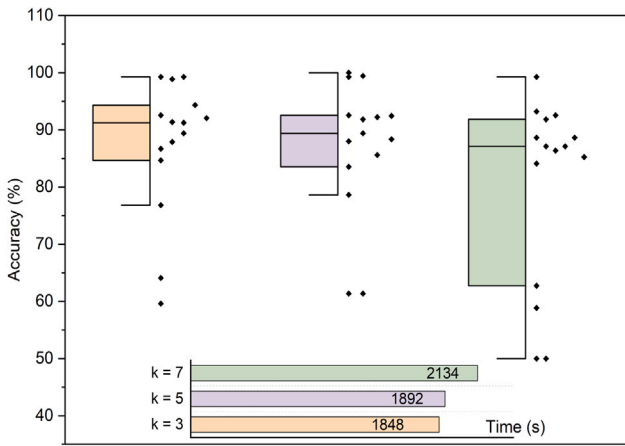


Fig. 5. Performance of the model based on the filter sizes using accuracy and time complexity as criteria. The diamond symbol represents the accuracy of each patient in this study.

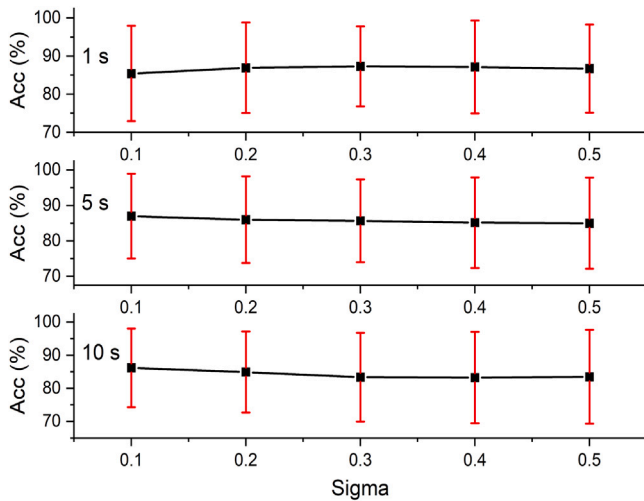


Fig. 6. Performance of GTF for different window lengths based on variable sigma. The error bars indicate the standard deviation of each sigma.

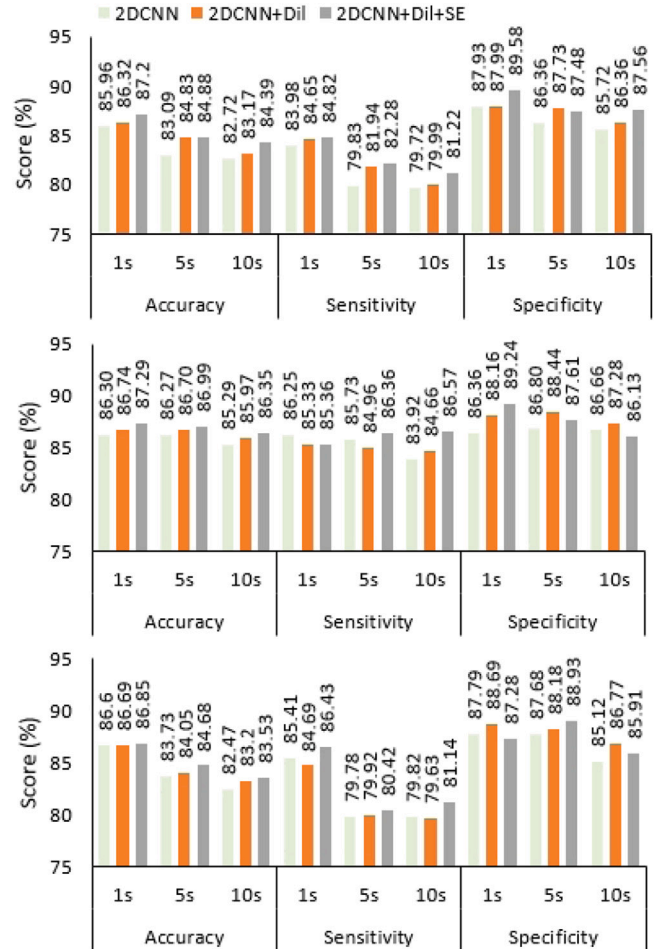


Fig. 7. Performance of the model based on the application of squeeze and excitation network across different window lengths for (a) Hann taper function, (b) Gaussian taper function, (c) without taper function.

3.3. Performance based on dilation convolution neural networks and squeeze and excitation networks

Fig. 7 illustrates the performance of the proposed dilated convolutional neural network with and without the SE block for three different window sizes, considering three metrics: accuracy, sensitivity, and specificity. The SE block application improves the model's performance across the three window lengths (WL = 1 s, WL = 5 s, and WL = 10 s). Initially, we examined the contribution of the dilation rate (2, 2) to 2DCNN before incorporating the squeeze and excitation networks. It can be observed that the addition of a dilation rate improves the performance of 2DCNN, as indicated by accuracy, sensitivity, and specificity scores. Since 2DCNN and dilated CNN have the same number of parameters but different receptive fields because of the dilation rate, we only discuss and compare the performance of dilated CNN with and without SE blocks. With the application of the Hann taper (Fig. 7a), the introduction of the SE blocks improves the accuracy of dilated CNN by 0.88%, 0.05%, and 1.22% for 1 s, 5 s, and 10 s, respectively. The sensitivity and specificity scores show similar improvements, particularly when WL = 1 s, achieving increases of 0.17% and 1.59%, respectively. In the case of the Gaussian taper function (Fig. 7b), the accuracy increased by 0.55% for 1 s, 0.29% for 5 s, and 0.38% for 10 s when the SE block was applied. Like the case with the Hann taper function, the application of the SE block alongside the Gaussian taper function increases the sensitivity of the dilated CNN by 0.03%, 1.40%, and 1.91% for window lengths of 1 s, 5 s, and 10 s respectively (Fig. 7b). However, SE blocks with GTF have little or no impact on the specificity of dilated CNN, especially for larger window sizes of WL = 5 s and WL = 10 s. Finally, Fig. 7c illustrates the model's performance without applying taper functions. The introduction of the SE blocks caused the accuracy to increase by 0.16%, 0.63%, and 0.33% for window lengths of 1 s, 5 s, and 10 s, respectively. Additionally, applying the SE block improves the model's sensitivity without taper functions by 1.74%, 0.5%, and 1.51% for WL = 1 s, WL = 5 s, and WL = 10 s, respectively.

3.4. Performance of the proposed DCSENet model of each patient through LOPOCV implementation

We assess the performance of our proposed DCSENet model for different window sizes (1 s, 5 s, and 10 s) with and without the application of taper functions (GTF and HTF) as shown in Table 3. The Kolmogorov–Smirnov (KS) test was also introduced to evaluate the statistical significance of the differences between training and test samples in LOPOCV implementation. We hypothesized that no significant difference exists between the training and test sets at a threshold of $0.01 < p \leq 0.05$ or $p > 0.05$. The KS test allows the model's performance to be checked against the distribution of the samples in training and test sets for each patient.

3.4.1. Performance with Hann taper function

Table 3 shows each patient's training and test samples. Patients Chb02, Chb07, Chb17, and Chb19 have the lowest test samples of 132, while Chb06 has the highest with 396 samples. When HTF is applied with the lowest window length (WL = 1 s), patients Chb01, Chb02, Chb03, Chb05, Chb07, Chb09, Chb10, Chb17, Chb18, Chb19 and Chb20 show good performance with a testing accuracy of 86.69%, 87.88%, 92.53%, 91.36%, 99.24%, 98.86%, 91.23%, 89.39%, 94.32%, 99.24% and 78.90%, respectively. These patients likewise demonstrated significant performances with WL of 5 s and 10 s respectively. Conversely, reduced model performances can be observed with patients Chb06, Chb08, and Chb16 with an average accuracy of 54.88%, 74.85%, and 63.79%, respectively across the three window lengths. Interestingly, the average p-values of 0.3685, 0.3273, 0.1616, and 0.4591 obtained for Chb01, Chb07, Chb10, and Chb18, respectively, are higher than the threshold value of 0.05, indicating no significant

difference between their train and the test sets. This is expected, as the test results for unseen samples of these patients achieved high performance. Similarly, patients Chb02, Chb03, Chb05, and Chb19 with p-values of 0.0177, 0.0353, 0.0139, and 0.0263, respectively show no significant difference between training and test samples at $0.01 < p \leq 0.05$. In contrast, the average p-values for low-performing patients (Chb06, Chb08, and Chb16) are lower than threshold $p = 0.05$, suggesting the variation in the distribution of their test samples to trained samples.

3.4.2. Performance with Gaussian taper function

The GTF shows similar performance to the HTF with slight differences. With the same patients, Chb01, Chb02, Chb03, Chb05, Chb07, Chb09, Chb10, Chb17, Chb18, Chb19, and Chb20, with an average accuracy of 85.39%, 85.86%, 91.99%, 92.12%, 98.73%, 90.05%, 91.02%, 89.14%, 93.56%, 99.75% and 91.48% across the three window lengths show better performance. Furthermore, patients Chb01, Chb02, Chb05, Chb07, and Chb18 had average p-values of 0.0917, 0.0905, 0.3202, 0.4173, and 0.2114, respectively, which exceed the threshold of 0.05. This suggests that these patients' test samples are being tested against train samples that have similar distributions. This results further confirm the reasons for these patients' high performance in the LOPOCV implementation. Conversely, patients Chb06, Chb08, and Chb16 again reported average p-values of 0.0003, 0.0000, and 0.0000, respectively, signifying a significant difference between their test and training samples distribution (Table 3).

3.4.3. Performance without taper function

As indicated in Table 3, the performance of the proposed model without taper functions (WTF) shows a similar trend to those involving Hann and Gaussian taper functions. Patients Chb01, Chb03, Chb07, and Chb18 show no significant difference between training and test samples at $p > 0.05$ while patients Chb06, Chb08, and Chb16 continue to indicate significant differences with reduced performance. In addition, the comparison using the overall average accuracy of the model with and without taper functions demonstrates the advantage of applying taper functions to minimize spectral leakage. The Hann taper function can be seen to improve the performance of WTF by an average accuracy increase of 0.35%, 0.20%, and 0.86% for WLs of 1 s, 5 s, and 10 s respectively. The comparison of WTF with GTF further illustrates the advantage of controlling the width of the window function by σ for any given window length. The result shows that GTF improves the performance of WTF by average accuracy of 0.44%, 2.31%, and 2.82% for WL = 1 s, WL = 5 s, and WL = 10 s, respectively. In addition, as the window lengths increased the performance of the WTF dropped significantly as compared to GTF. This is anticipated as the initial ablation study conducted on GTF ensures that appropriate σ values are selected for each window length.

3.5. Effects of taper functions

Fig. 8 indicates the distribution of the samples with and without taper functions using the low (WL = 1 s) and high (WL = 10 s) window lengths. The figure shows the interpolation of the power spectrum in the time-frequency domain (refer to Fig. 3f for color bar). The application of HTF with WL = 1 s indicates the removal of some edge effects in the spectrogram with some low power spectrum visualization in-between the high power spectrum (Fig. 8a, left) as compared without applying the taper function (Fig. 8c, left). Similarly, the application of GTF shows some low spectrum around the edges. As the t-SNE plots indicate, the model (DCSENet) could differentiate between the normal and seizure events when WL = 1 s for the three conditions. However, when the window length is increased to 10 s, lower power spectrums (in decibels) can be observed in the seizure spectrogram for the Hann window (Fig. 8a, right). As a result, the t-SNE shows that some normal samples are partially diffused into the seizure samples. Likewise, the

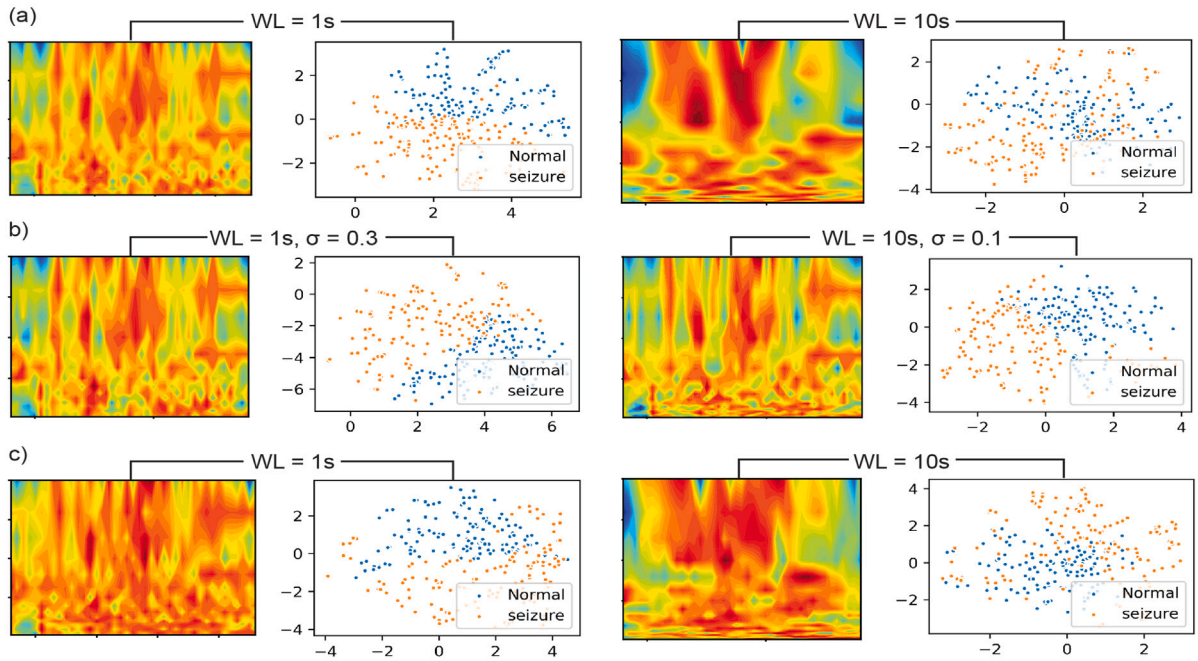


Fig. 8. Effects of window function on the distribution of samples using t-distributed stochastic neighbor embedding (t-SNE) on the spectrogram (frequency on vertical axis, time on horizontal axis) of typical seizure event of patient Chb01 (a) Hann taper function using WL of 1 s and 10 s (b) Gaussian taper function at WL = 1 s, $\sigma = 0.3$ and WL = 10 s, $\sigma = 0.1$ (c) without the taper function at WL = 1 s and WL = 10 s.

increase in window size to $W = 10$ s without taper function (Fig. 8c, right) shows the power spectrum diffusing into each other with the t-SNE plot unable to discriminate between normal and seizure samples effectively. Moreover, the t-SNE distribution further explains the reason for the performance of the HTF and WTF when smaller and larger window sizes of 1 s and 10 s, respectively are used, as shown in Table 3. In the case of HTF, the average accuracy across the patients is $87.20 \pm 11.48\%$ and $84.39 \pm 13.52\%$ for WL = 1 s and $W = 10$ s, respectively. However, WTF, shows lower performance with an average accuracy of $86.85 \pm 11.56\%$ and $83.53 \pm 14.55\%$ for WL = 1 s and WL = 10 s, respectively (Table 3). The results agree with t-SNE distribution plots for WTF as shown in Fig. 8c.

Moreover, the application of GTF at WL = 10 s with $\sigma = 0.1$ and WL = 1 s with $\sigma = 0.3$ as shown in (Fig. 8b, right) demonstrates the advantage of controlling the width of the window function. At WL = 1 s, a similar trend in power spectrums and t-SNE distribution can be observed for both GTF and HTF (Fig. 8 a and b, left). However, at WL = 10 s for GTF, we observe a few spectra with low intensity within the high-power spectral region. This can be attributed to the effective window control on GFT by using $\sigma = 0.1$ for WL of 10 s. Moreover, seizure and normal samples are better discriminated by GTF at WL = 10 s than HTF of the same window length (Fig. 8 a and b, right). Furthermore, the performance of GTF based on t-SNE distribution justifies the reason for the average performance of the model reported for GFT in Table 3. With WL = 1 s for GTF, the average accuracy achieved is $87.29 \pm 10.48\%$ and $86.35 \pm 12.07\%$ when WL = 10 s (Table 3).

4. Discussion

The development of machine-learning models for epileptic seizure detection for individual patients has received more attention recently. The choice of patient-specific models is due to the high inter-subject variability of EEG recordings, which often results in low performance of models in patient-independent tasks [31]. However, patient-specific models suffer from generalization problems and cannot reproduce similar results on unseen patient datasets. Patient-independent seizure detection based on feature selection has been introduced [2,31]. In

addition, leave-one-out cross-validation (LOOCV) has been successfully applied in seizure prediction for patient-specific approaches [8,15]. However, implementing a patient-independent algorithm, especially with the full utilization of LOPOCV for ESD, has yet to receive much attention. Moreover, LOPOCV can give insight into the model's generalization ability, ensuring its suitability for different patients. Although, EEG signals are complex and non-stationary, accompanied by noise and other artifacts [2]. SFTF can be used to preprocess the signal, thereby allowing deep-learning models to extract informative features more effectively [8]. Furthermore, STFT can provide better time-frequency resolution with a clinical information channel that neurologists can easily interpret. Therefore, introducing taper functions of variable length is essential to improved time-frequency resolution during spectrogram construction of EEG signals for epileptic seizure detection.

4.1. Differences and similarities among patients' samples

The Kolmogorov–Smirnov test was applied to understand the differences and similarities between the distribution of training and test samples of the patients. KS test is a non-parametric method to determine if two samples originate from the same distribution [32]. This makes it particularly suitable for EEG signals, which are non-linear and non-Gaussian in nature. Additionally, the KS test is specifically designed for continuous distributions, which is essential for representing time-frequency data distribution. The KS test was successfully applied by [33] to evaluate the significant difference between extracted Hurst exponent features of normal EEG recordings and seizure-free intervals from healthy subjects and seizure patients, respectively. The KS test likewise enhances the feature selection process in [34]. However, in this study, we applied the KS test to identify the significant difference between training and test samples, which allowed us to better understand the variation in patient performance under the LOPOCV approach. With and without the application of taper functions, patients such as Chb01, Chb02, Chb03, Chb05, Chb07, Chb09, Chb10, Chb18, and Chb19 show better performance, with their p-values confirming significant similarity between their test samples and trained samples from different patients (Table 3). The similarity in sample distribution

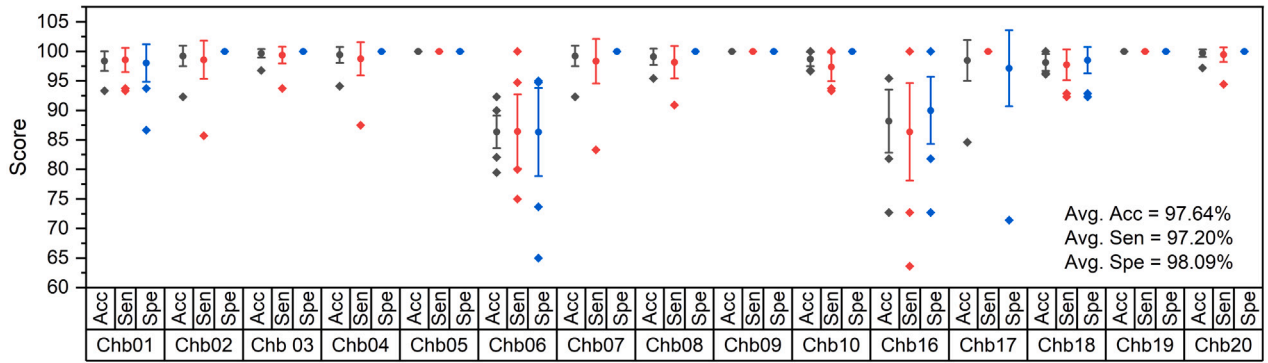


Fig. 9. Patient-specific spectrogram classification using DCSENNets through 10-fold cross-validation. The error bars indicate the standard deviation values for each patient, suggesting individual pattern variation.

of these patients, as indicated by KS p-values, suggests their ability to be combined in building a generalized and reliable model for seizure detection.

In contrast, patients Chb06 and Chb16 achieved lower testing accuracy with or without applying taper functions, as confirmed by $p < 0.01$. This shows a significant difference between the training and testing samples for these two patients. The reported high performance for certain patients and low performance for others may be due to age differences, gender distribution, or recording conditions. Aged patients, such as Chb03, Chb04, Chb07, Chb18, and Chb19, ages 14, 22, 14.5, 18, and 19 years old, can be more coordinated during data acquisition and less noise recorded; hence, these patients have high model performance. However, the low performance of patients Chb06 and Chb16 may be due to low age differences or recording conditions. As a result, we performed patient-specific classification using a 10-fold cross-validation (CV) for each patient to gain a better understanding of the within-patient sample distribution. As shown in Fig. 9, all the patients almost achieved perfect classification as indicated by their accuracy, sensitivity, and specificity scores, except for patients Chb06 and Chb16 with low performances. The deviation in performance of patients Chb06 and Chb16 for 10-fold CV, as indicated by the error bar, justifies their low performances in LOPOCV implementation. In addition, the performance can also be attributed to amplitude difference and noisy recording, especially the EEG of patient Chb06, an infant (1.5 years) [35].

4.2. Impact of squeeze and excitation networks

Seizures are broadly classified into partial and generalized seizures based on the epileptogenic zone within the brain. Identifying, analyzing, and interpreting signals from channels in this region are crucial for accurate diagnosis. Paying attention to informative channels will help distinguish between neurophysiological profiles of patients. The SE block can enhance the model's performance by giving higher excitation scores to the channels with relevant information [29] in the feature maps. As indicated in Fig. 7, adding SE blocks improves the performance of the proposed model through accuracy, sensitivity, and specificity. Higher specificity values than sensitivity for conventional 2DCNN and dilated CNN demonstrate the constant trade-off between sensitivity and specificity. However, the inclusion of SE blocks potentially improves both sensitivity and specificity, especially for WL of 1 s. Therefore, the addition of SE blocks demonstrated potential improvement in feature calibration by dynamically adjusting focus to relevant features across multiple patients. This allows the proposed model to pay attention to the informative channels. Thereby enhancing the overall accuracy of the proposed patient-independent seizure detection algorithm.

4.3. Time-frequency resolution of spectrograms

High time-frequency resolution of the spectrogram is essential to provide clinical interpretation of individual channels in EEG signals. Moreover, deep learning models can achieve optimum performance based on the features learned from the spectrogram. There is a contention between time and frequency resolution. Increasing the window length enhances the frequency resolution but reduces temporal clarity, whereas decreasing the window length results in spectrograms with better time resolution at the cost of frequency resolution [9,38]. Therefore, we experimented with different window sizes to identify the optimum window length. Fig. 8 illustrates the performance of the proposed model for the lowest window length (WL = 1 s) and highest window length (WL = 10 s) considered in this study. When WL = 1 s, there is a balance between time and frequency resolutions, as indicated by the power spectrum distribution. Hence, the model could differentiate between the samples from non-seizure and seizure intervals, as shown in Fig. 8(a,b left). By increasing the window length to 10 s, the spectrogram shows better frequency resolution at the expense of time resolution, as shown in Fig. 8(a right). Similar results are obtained in Fig. 8c, except in Fig. 8b for GTF, where the model could still differentiate the samples. The outcome of GTF can be attributed to the small width of GTF with $\sigma = 0.1$ (Fig. 2) that forces the spectrogram for WL = 10 s to resemble that of WL = 1 s at $\sigma = 0.3$. Therefore, care must be taken when selecting the window length to provide a balance between frequency and time resolution for the spectrograms. Moreover, ensuring a balance between temporal and spectral features in the spectrogram images enables proposed DCSENNets to learn robust features for efficient seizure detection.

4.4. Explainability of the deep learning model performance

We utilized the Grad-CAM deep explainer to comprehend the crucial areas of the spectrogram that the model prioritizes when making classification decisions for seizures and normal conditions. Deep visual explainer results of the model with and without the application of the taper function were compared, highlighting the regions in the spectrogram that are prioritized for classification. The visual explanation shows the application of the Hann taper allows the model to focus on fewer spectrums than without the taper function while making classification decisions with better probability (Fig. 10). With this specificity, neurologists can easily understand the deep learning model's decision-making process and efficiently identify the epileptogenic zone.

Furthermore, Fig. 11 explains the results of the patient-independent (Table 3) approach using a confusion matrix. The confusion matrix represented by true positive (TP), false negative (FN), false positive (FP), and true negative (TN) scores detailed the reasons for each patient's performance. For optimal performance, we aim to achieve 50% TP and 50% TN, with 0% FN and 0% FP. While most patients

Table 4

Performance comparison of proposed method with other deep learning spectrogram seizure detection algorithms.

Work	Method	No of patients	Explainability	Approach	Validation method	Acc (%)	Sen (%)	Spec (%)
[36]	Spectrogram + Inception-v3	4	N/A	PI ₂	4-fold CV	88.34	–	–
[7]	Spectrogram encrypted by Chaotic Baker Map + CNN (Googlenet)	23	N/A	PI	70% holdout	84.72	88.89	–
[37]	Spectrogram + 2DCNN	8	N/A	PS	70% holdout	77.57	75.59	79.54
[10]	Rhythmicity spectrogram + CNN	4	N/A	PI	80% holdout	61.00	59.00	–
[30]	Spectrogram + three layers CNN	8	N/A	PS	70% holdout	92.49	93.18	92.71
Ours	Enhanced STFT Spectrogram Visualization + DCSENNets	15	Deep XAI	PS	10-fold CV	97.64	97.20	98.09
				PI₁	LOPOCV	87.29	85.36	89.24
				PI₂	10-fold CV	94.48	93.50	95.46

PI = patient-independent, PS = patient-specific, PI₁ = patient-independent using LOPOCV and PI₂ = patient-independent using CV (cross-validation), N/A = Not available and deep explainable AI (Deep XAI).

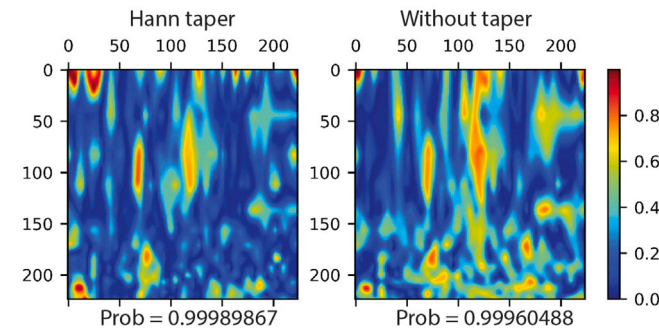


Fig. 10. Grad CAM visual explainability of the proposed deep learning model on spectrograms of Fig. 8a (left) and Fig. 8c (right). The figures indicate the relevant areas the model makes decisions on with their probability scores.

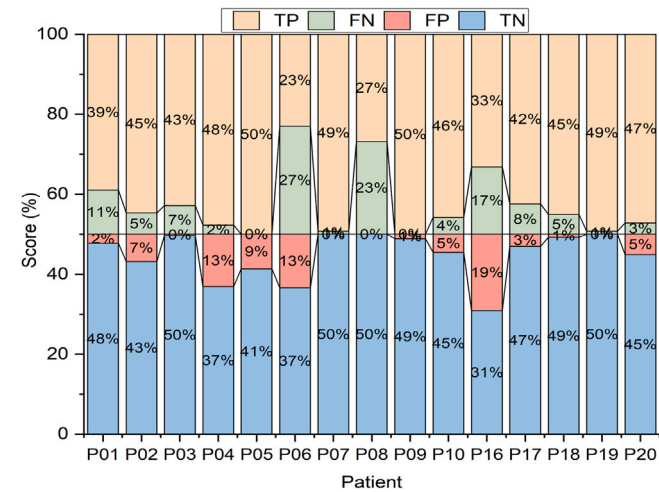


Fig. 11. Performance of individual patient based on the confusion matrix. TP, FN, FP, and TN represent True Positive, False Negative, False Positive, and True Negative, respectively.

nearly achieve this performance, there are instances of misclassification in some patients, leading to an increase in FN and FP scores. Patients Chb06 and Chb16 show higher FN and FP scores than any other patient. A false negative score of 27% achieved for patient Chb06 shows that more of the seizure spectrograms of this patient are misclassified as normal spectrograms. This implies a close similarity in normal and seizure patterns for patient Chb06. Additionally, patient Chb16 displays a high FN and a correspondingly high FP, indicating that the model cannot accurately classify certain normal and seizure spectrograms

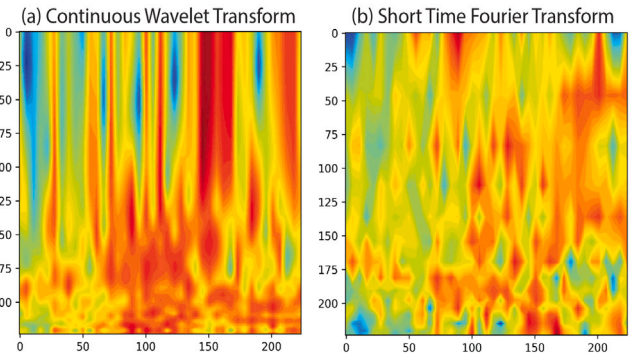


Fig. 12. Scalogram and spectrogram representation of continuous wavelet transform (CWT) and short time Fourier Transform (STFT). The Scalogram of CWT was constructed using a Morlet wavelet.

for this patient. This further explains the variability within individual patient datasets resulting in their performances, as shown in Fig. 9.

4.5. Performance comparison of proposed STFT with CWT

Continuous wavelet transform (CWT) is another important tool for capturing time-frequency components of EEG recordings through signal-to-image transformation. Hence, we compare the performance of the proposed enhanced STFT spectrogram with a CWT scalogram that uses Morlet wavelet as shown in Fig. 12. It can be observed that STFT provides a uniform resolution across frequencies, whereas CWT spreads out frequency-related information. Moreover, higher frequency resolution can be observed in the CWT at the expense of time resolution, with an average classification accuracy of $87.32 \pm 13.99\%$ (Fig. 13). However, the proposed STFT approach ensures a balanced time-frequency resolution with an average seizure classification accuracy of $87.29 \pm 10.48\%$. The results agreed with the conclusion from [39] that STFT and CWT indicate similar performance across the eight-fold cross-validation. Additionally, CWT is computationally intensive and is sensitive to noise as signals are convolved with wavelets at multiple scales. Furthermore, the interpretation of the generated images constructed on EEG signals is essential for neurologists. STFT outputs are easier to interpret than CWT, as they yield a constant time-frequency resolution.

4.6. Comparison with previous methods and studies

Initially, we compare the performance of our proposed DCSENNets model with other cutting-edge models for image classification. We use the models (ResNet50, VGG16, and InceptionV3) on our generated spectrogram images for ESD to ensure a fair comparison. The results in Fig. 14 show that our proposed model achieved better performance with an average accuracy of 87.20% compared to an average accuracy

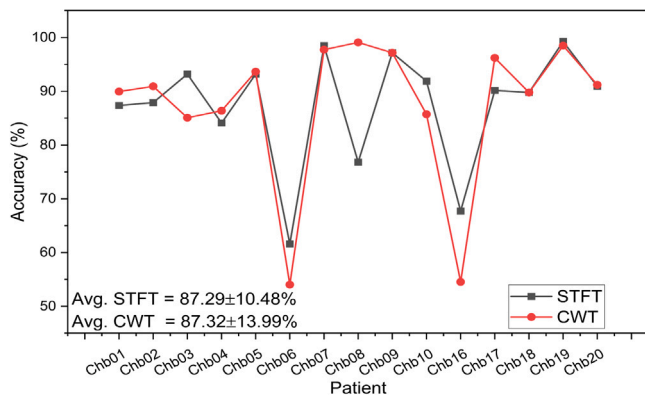


Fig. 13. Performance comparison of classification outcome of proposed STFT with CWT across the patients.

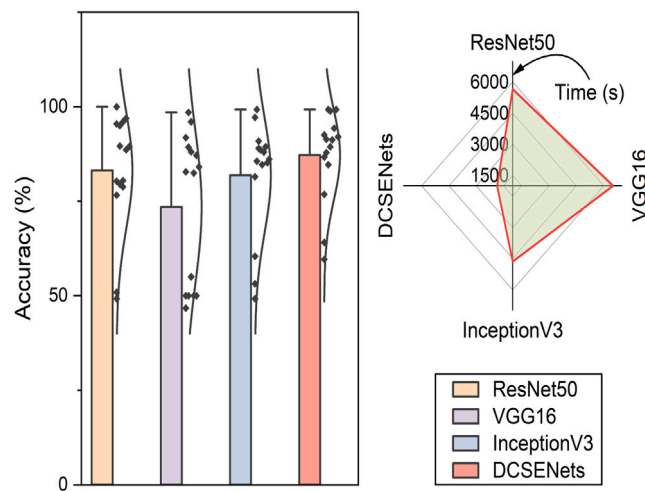


Fig. 14. Performance comparison of proposed DCSEnets with state-of-the-art models. The comparison is based on the accuracy and computational demand of each model structure where time is measured in seconds. The diamond symbols indicate the accuracy for each of the 15 patients and the bars show the standard deviation.

of 83.14%, 73.45%, and 81.88% obtained using ResNet50, VGG16, and InceptionV3, respectively. In addition, the proposed model took a relatively short duration of 1848 s to loop through the 15 patients, whereas ResNet50, VGG16, and InceptionV3 needed more than 4500 s to complete the patient-independent classification running through the same 20 epochs. We attribute our proposed model's computational efficiency to its relatively simple structure with fewer parameters. The standard deviation bars also show that DCSEnets are better at generalization. This demonstrates the robustness of DCSEnets in adaptively scaling the features and modeling interdependencies between channels across different patient datasets.

In addition, Table 4 shows how our enhanced STFT spectrogram construction method compares against other deep-learning seizure classification methods that use spectrogram images. We performed a similar patient-independent technique using 10-fold CV (PI_2) for all the combined patient data, achieving an accuracy, sensitivity, and specificity of 94.48%, 93.50%, and 95.46%, respectively, which is better than those in [7,36]. Furthermore, the average accuracy for LOPOCV in this study (PI_1) is 87.29% when GTF with $WL = 1$ and $\sigma = 0.3$ was used. This is higher than the average accuracy of 61% reported by [10], which used four patients and 15 EEG channels. With a patient-specific approach, the proposed algorithm demonstrated superior performance to the works of [30,37] with average accuracy,

sensitivity, and specificity of 97.64%, 97.20%, and 98.09% respectively. Although spectrogram and phase space reconstruction images are used in [30], we only considered the spectrogram results for a fair comparison in this study. Finally, unlike other comparative studies, our approach incorporates explainable AI to understand the deep learning model's decision-making process in identifying essential parts of the spectrogram.

4.7. Limitations and future work

The proposed algorithm has demonstrated a high potential for patient-independent seizure detection. However, addressing certain constraints can further improve its effectiveness. The algorithm exhibits instances of misclassification, particularly within specific age groups, resulting in low classification accuracy for certain patients in these demographics. Although some patients within this age group show significant performance, lack of coordination or noisy recordings could have resulted in low performance among identified patients. Moreover, we have randomly selected the normal interval from the interictal state without regard for recordings made under unfavorable device settings. Despite the implementation of preprocessing procedures, randomly including noisy recordings or normal-like seizure patterns can impact model performance, especially for pediatric patients with clonic and tonic seizures. This emphasizes the need for additional information, especially for pediatric patients whose seizure patterns are not easily distinguishable from normal activity. Multimodal data integration of physiological data such as ECG (electrocardiogram), EMG (electromyography), and pulse oximetry with EEG recordings could improve the classification output of challenging cases of pediatric patients. Additionally, we will integrate sequential models such as temporal convolutional networks (TCNs) or LSTM with the STFT-DCSEnets model in our future work to enhance the analysis by capturing temporal dependencies alongside spectral information.

5. Conclusion

This study introduces a generalized approach for epilepsy detection by leveraging enhanced STFT spectrogram visualized images through the proposed DCSEnets model. Enhanced spectrograms are clinically important for neurologists to identify seizure patterns easily. Therefore, we introduced taper functions to control the spectral leakage and increase the precision of seizure identification. The taper functions help to smoothen the edges of the window to reduce discontinuity at the boundaries of the windowed channels of the EEG signals during spectrogram construction. Moreover, preprocessing EEG signals in this way enables effective feature extraction and channel-wise calibration using the proposed DCSEnets. Thus, the proposed DCSEnets effectively distinguish between non-seizure and seizure channels, especially with a taper function implementation. Furthermore, including a visual explainer creates an interpretable system for neurologists to understand the DCSEnets model's decision-making process. In addition, the adjusted time-frequency resolution through variable window lengths enhances the visualization of spectrograms. This introduces flexibility to diagnostic tools, allowing neurologists to optimize spectrograms for various seizure types and individual patient conditions. Finally, the enhanced spectrogram visualization with a deep explainer can be integrated into EEG real-time monitoring systems to dynamically identify epileptogenic zones for a more holistic diagnosis.

CRedit authorship contribution statement

Sunday Timothy Aboyeji: Writing – original draft, Visualization, Validation, Formal analysis. **Ijaz Ahmad:** Methodology, Formal analysis. **Xin Wang:** Writing – review & editing, Supervision, Funding acquisition. **Yan Chen:** Funding acquisition, Conceptualization. **Chen Yao:** Investigation. **Guanglin Li:** Supervision, Funding acquisition,

Conceptualization. **Michael Chi Fai Tong:** Supervision, Conceptualization. **Alice K.Y. Siu:** Supervision, Conceptualization. **Guoru Zhao:** Supervision. **Shixiong Chen:** Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declared that they have no conflicts of interest to this work. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

References

- [1] A. Shoeibi, P. Moridian, M. Khodatars, N. Ghassemi, M. Jafari, R. Alizadehsani, Y. Kong, J.M. Gorriz, J. Ramirez, A. Khosravi, et al., An overview of deep learning techniques for epileptic seizures detection and prediction based on neuroimaging modalities: Methods, challenges, and future works, *Comput. Biol. Med.* 149 (2022) 106053.
- [2] S. Yang, B. Li, Y. Zhang, M. Duan, S. Liu, Y. Zhang, X. Feng, R. Tan, L. Huang, F. Zhou, Selection of features for patient-independent detection of seizure events using scalp EEG signals, *Comput. Biol. Med.* 119 (2020) 103671.
- [3] I. Ahmad, X. Wang, D. Javeed, P. Kumar, O.W. Samuel, S. Chen, A hybrid deep learning approach for epileptic seizure detection in EEG signals, *IEEE J. Biomed. Health Inform.* (2023).
- [4] R.I. Halawa, S.M. Youssef, M.N. Elagamy, An efficient hybrid model for patient-independent seizure prediction using deep learning, *Appl. Sci.* 12 (11) (2022) 5516.
- [5] X. Wang, Y. Wang, D. Liu, Y. Wang, Z. Wang, Automated recognition of epilepsy from EEG signals using a combining space-time algorithm of CNN-LSTM, *Sci. Rep.* 13 (1) (2023) 14876.
- [6] N.A. Samee, N.F. Mahmoud, E.A. Aldahri, A. Rafiq, M.S.A. Muthanna, I. Ahmad, RNN and BiLSTM fusion for accurate automatic epileptic seizure diagnosis using EEG signals, *Life* 12 (12) (2022) 1946.
- [7] A.A.E. Shoka, M.M. Dessouky, A. El-Sayed, E.E.-D. Hemdan, An efficient CNN based epileptic seizures detection framework using encrypted EEG signals for secure telemedicine applications, *Alex. Eng. J.* 65 (2023) 399–412.
- [8] X. Lu, A. Wen, L. Sun, H. Wang, Y. Guo, Y. Ren, An epileptic seizure prediction method based on CBAM-3D CNN-LSTM model, *IEEE J. Transl. Eng. Health Med.* (2023).
- [9] J.S. Ra, T. Li, et al., A novel epileptic seizure prediction method based on synchroextracting transform and 1-dimensional convolutional neural network, *Comput. Methods Programs Biomed.* 240 (2023) 107678.
- [10] P. Handa, N. Goel, Epileptic seizure detection using rhythmicity spectrogram and cross-patient test set, in: 2021 8th International Conference on Signal Processing and Integrated Networks, SPIN, IEEE, 2021, pp. 898–902.
- [11] A. Keil, E.M. Bernat, M.X. Cohen, M. Ding, M. Fabiani, G. Gratton, E.S. Kappenman, E. Maris, K.E. Mathewson, R.T. Ward, et al., Recommendations and publication guidelines for studies using frequency domain and time-frequency domain analyses of neural time series, *Psychophysiology* 59 (5) (2022) e14052.
- [12] S. Tang, C. Jing, Y. Jiang, K. Yang, Z. Huang, H. Wu, C. Cui, S. Shi, X. Ye, H. Tian, et al., The effect of image resolution on convolutional neural networks in breast ultrasound, *Heliyon* 9 (8) (2023).
- [13] N. Ke, T. Lin, Z. Lin, Channel-weighted squeeze-and-excitation networks for epileptic seizure detection, in: 2021 IEEE 33rd International Conference on Tools with Artificial Intelligence, ICTAI, IEEE, 2021, pp. 666–673.
- [14] A.L. Goldberger, L.A. Amaral, L. Glass, J.M. Hausdorff, P.C. Ivanov, R.G. Mark, J.E. Mietus, G.B. Moody, C.-K. Peng, H.E. Stanley, PhysioBank, PhysioToolkit, and PhysioNet: components of a new research resource for complex physiologic signals, *Circulation* 101 (23) (2000) e215–e220.
- [15] H. Daoud, M.A. Bayoumi, Efficient epileptic seizure prediction based on deep learning, *IEEE Trans. Biomed. Circuits Syst.* 13 (5) (2019) 804–813.
- [16] N. Pielawski, C. Wählby, Introducing Hann windows for reducing edge-effects in patch-based image segmentation, *PLoS One* 15 (3) (2020) e0229839.
- [17] S. Braun, Identification, fourier-based methods, 2001.
- [18] Y. Jin, Z.Y. Hao, Gaussian window of optimal time-frequency resolution in numerical implementation of short-time fourier transform, *Appl. Mech. Mater.* 48 (2011) 555–560.
- [19] S. Shimauchi, S. Kudo, Y. Koizumi, K. Furuya, On relationships between amplitude and phase of short-time Fourier transform, in: 2017 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP, IEEE, 2017, pp. 676–680.
- [20] N. Goel, J. Singh, Analysis of kaiser and gaussian window functions in the fractional fourier transform domain and its application, *Iran. J. Sci. Technol. Trans. Electr. Eng.* 43 (2) (2019) 181–188.
- [21] F.J. Harris, On the use of windows for harmonic analysis with the discrete Fourier transform, *Proc. IEEE* 66 (1) (1978) 51–83.
- [22] N. Kehtarnavaz, Frequency domain processing, *Digit. Signal Process. Syst. Des.* 1 (2008) 175–196.
- [23] X. Yao, X. Wang, S.-H. Wang, Y.-D. Zhang, A comprehensive survey on convolutional neural network in medical image analysis, *Multimedia Tools Appl.* 81 (29) (2022) 41361–41405.
- [24] Q. Wang, Z. Yuan, Q. Du, X. Li, GETNET: A general end-to-end 2-D CNN framework for hyperspectral image change detection, *IEEE Trans. Geosci. Remote Sens.* 57 (1) (2018) 3–13.
- [25] A. Mohan, V. Meenakshi Sundaram, V3o2: hybrid deep learning model for hyperspectral image classification using vanilla-3D and octave-2D convolution, *J. Real-Time Image Process.* 18 (5) (2021) 1681–1695.
- [26] C. Zhang, J. Feng, C. Hu, Z. Liu, L. Cheng, Y. Zhou, An intelligent fault diagnosis method of rolling bearing under variable working loads using 1-D stacked dilated convolutional neural network, *IEEE Access* 8 (2020) 63027–63042.
- [27] X. Zhang, Y. Zou, W. Shi, Dilated convolution neural network with LeakyReLU for environmental sound classification, in: 2017 22nd International Conference on Digital Signal Processing, DSP, IEEE, 2017, pp. 1–5.
- [28] J. Hu, L. Shen, G. Sun, Squeeze-and-excitation networks, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 7132–7141.
- [29] S.K. Roy, S.R. Dubey, S. Chatterjee, B. Baran Chaudhuri, FuSENet: fused squeeze-and-excitation network for spectral-spatial hyperspectral image classification, *IET Image Process.* 14 (8) (2020) 1653–1661.
- [30] F.E. Ibrahim, H.M. Emara, W. El-Shafai, M. Elwekeil, M. Rihan, I.M. Eldokany, T.E. Taha, A.S. El-Fishawy, E.-S.M. El-Rabaie, E. Abdellatif, et al., Deep-learning-based seizure detection and prediction from electroencephalography signals, *Int. J. Numer. Methods Biomed. Eng.* 38 (6) (2022) e3573.
- [31] L. Abou-Abbas, K. Henni, I. Jemal, A. Mitiche, N. Mezghani, Patient-independent epileptic seizure detection by stable feature selection, *Expert Syst. Appl.* 232 (2023) 120585.
- [32] B. Zhang, R. Chen, Nonlinear time series clustering based on Kolmogorov-Smirnov 2D statistic, *J. Classification* 35 (2018) 394–421.
- [33] S. Lahmiri, An accurate system to distinguish between normal and abnormal electroencephalogram records with epileptic seizure free intervals, *Biomed. Signal Process. Control* 40 (2018) 312–317.
- [34] H. Al-Hadeethi, S. Abdulla, M. Diykh, J.H. Green, Determinant of covariance matrix model coupled with adaboost classification algorithm for EEG seizure detection, *Diagnostics* 12 (1) (2021) 74.
- [35] L. Wei, C. Mooney, Pediatric and adolescent seizure detection: A machine learning approach exploring the influence of age and sex in electroencephalogram analysis, *BioMedInformatics* 4 (1) (2024) 796–810.
- [36] E. Gupta, M. Gupta, R.A. Sachdeva, P. Handa, N. Goel, Automatic seizure detection using rhythmicity spectrograms and inception-v3 architecture, in: 2023 10th International Conference on Signal Processing and Integrated Networks, SPIN, IEEE, 2023, pp. 131–136.
- [37] G.C. Jana, R. Sharma, A. Agrawal, A 1D-CNN-spectrogram based approach for seizure detection from EEG signal, *Procedia Comput. Sci.* 167 (2020) 403–412.
- [38] A. Zabidi, W. Mansor, Y. Lee, C.C.W. Fadzal, Short-time Fourier transform analysis of EEG signal generated during imagined writing, in: 2012 International Conference on System Engineering and Technology, ICSET, IEEE, 2012, pp. 1–4.
- [39] M. Varli, H. Yilmaz, Multiple classification of EEG signals and epileptic seizure diagnosis with combined deep learning, *J. Comput. Sci.* 67 (2023) 101943.